

MDI3003
ADVANCED PREDICTIVE ANALYTICS
LABORATORY MANUAL - EXPERIMENT 08
Agricultural Predictive Analytics - Crop Yield
Prediction with a Guided Crop-Label Classification
Extension

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GITHUB REPO LINK :

https://github.com/srijithramesh24/24MID0009_Lab08_Agricultural-Predictive-Analytics

ABSTRACT

Agricultural yield prediction is an important predictive analytics problem because crop productivity varies across geographical regions and over time. This experiment develops a machine-learning pipeline for predicting rice yield at the district–season–year level using historical agricultural observations from India. The study follows a chronological evaluation design in which earlier observations are used for training, the following two years are used for validation, and the final two observed years are retained as an unseen test set.

Four regression approaches are compared: a median baseline, Ridge regression with a temporal trend, a Decision Tree regressor, and a Random Forest regressor. Categorical variables are one-hot encoded and the year variable is standardized within a training-fitted preprocessing pipeline. Model selection is performed using validation Mean Absolute Error (MAE), while RMSE and R^2 are used as secondary metrics.

The Ridge Trend model achieved the best validation performance with an MAE of **0.4282 t/ha** and was selected before opening the test set. On the locked test set, the selected model achieved an MAE of **0.4332 t/ha**, RMSE of **0.5707 t/ha**, and R^2 of **0.7596**, compared with a test MAE of **1.0095 t/ha** for the median baseline. The results demonstrate the usefulness of a temporal trend model for this defined historical prediction task while also highlighting the limitations caused by the absence of detailed weather, soil, irrigation and crop-variety information.

Keywords: Agricultural Predictive Analytics, Rice Yield Prediction, Ridge Regression, Random Forest, Decision Tree, Time-based Validation, Machine Learning, India

AIM

To develop, compare and evaluate reproducible machine-learning models for predicting district-level rice yield using historical agricultural data while maintaining chronological evaluation and preventing target/data leakage.

INTRODUCTION

Agriculture is strongly influenced by geographical, seasonal and temporal factors. Crop yield can vary considerably between districts and across different agricultural years due to differences in environmental conditions, agricultural practices and other regional characteristics. Predictive analytics can be used to learn patterns from historical observations and estimate crop yield for later observations.

In this experiment, the prediction problem is formulated around **rice yield prediction at the district–season–harvest-year level**. The objective is not to recommend the best crop or provide an agronomic prescription, but to evaluate whether machine-learning models can predict later observed rice yields using a restricted set of predictors.

The laboratory manual specifies that the core predictors are **state, district, season and harvest year**, while production and other target-derived variables are excluded. This is important because production and harvested area can be used to reconstruct yield and therefore would introduce target leakage. The manual also requires chronological evaluation so that future years are not used during model development.

The experiment compares simple and nonlinear predictive models under the same train-validation-test protocol. The final test set remains locked until model selection is completed.

1. Problem Formulation

1.1 Objective

The objective of this experiment is to develop and evaluate machine learning models for predicting **rice yield in tonnes per hectare (t/ha)** using district-level agricultural observations. The prediction unit is one **district–season–harvest-year observation**.

The models use only information intended to be available before the target harvest:

- State
- District
- Season
- Harvest year

The target variable is `yield_t_ha`.

The experiment compares a simple **median baseline**, **Ridge regression with a year trend**, **Decision Tree regression**, and **Random Forest regression**. The models are evaluated using a chronological split so that later years are not used for training earlier-year predictions.

The primary model-selection metric is **Mean Absolute Error (MAE)**. RMSE and R^2 are used as secondary evaluation measures.

1.2 Prediction Contract

Item	Description
Prediction unit	District–season–harvest-year
Crop	Rice
Target	Rice yield
Target unit	tonnes/hectare (t/ha)
Predictors	State, district, season, year
Primary metric	MAE
Secondary metrics	RMSE, R^2
Training period	1990–2011
Validation period	2012–2013
Test period	2014–2015
Intended use	Regional planning/student predictive analysis

Production and harvested-area variables were not used as predictors because production and area can directly reconstruct yield and would therefore cause target leakage.

2. [Dataset and Data Preparation](#)

The regression dataset contains district-level agricultural observations covering **1990–2015**. The experiment uses the Rice-only scope and the canonical variables required by the laboratory manual.

The final predictor whitelist was:

- state
- district
- season
- year
- The target was:
- yield_t_ha

A unique row_id was retained to verify split membership and reproducibility.

The dataset was checked for:

- Missing targets
- Invalid yield values
- Invalid years
- Missing state/district/season values
- Duplicate agricultural keys
- Duplicate row IDs
- Chronological split consistency

The resulting dataset contains **15500 observations**, with **620 observations in each of the validation and test sets**.

3. [Train, Validation and Test Split](#)

The experiment follows the chronological evaluation procedure as specified in the given lab. manual.

The latest two observed years were reserved for the final test:

- 2014
- 2015

The preceding two years were used for validation:

- 2012
- 2013

All earlier years were used for training:

- 1990–2011

This prevents future-year information from entering model training.

Split Summary

Split	Years	Rows	Purpose
Training	1990–2011	13,640	Model fitting
Validation	2012–2013	620	Model selection
Test	2014–2015	620	Final evaluation

4. Exploratory Data Analysis

EDA was performed using the training data before model selection.

4.1 Training Yield Distribution

The training yield distribution was examined to understand the spread of rice yields and identify the presence of unusually low or high observations.

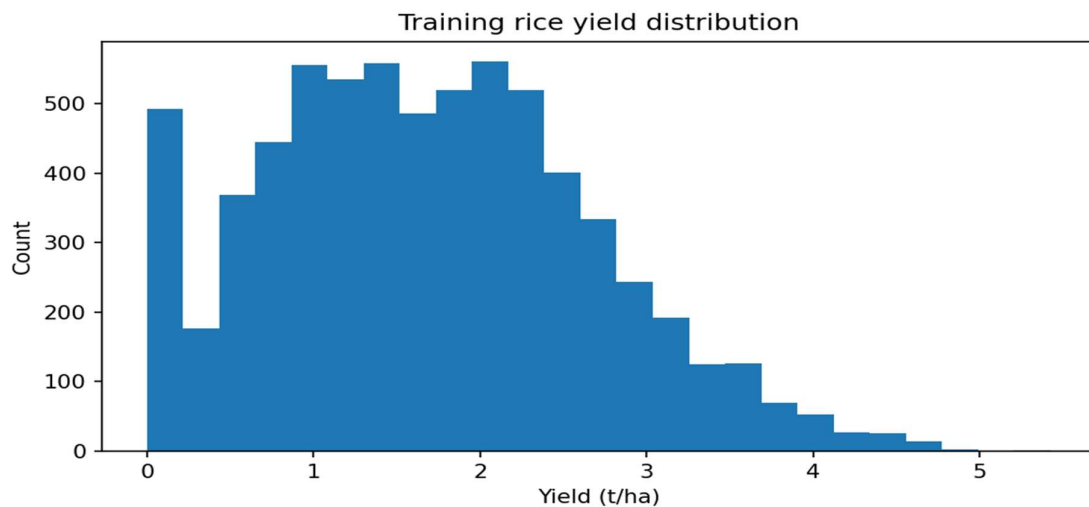


Figure: training_yield_distribution.png

Interpretation ; The training yield distribution shows considerable variation between district-level observations. This indicates that a single constant prediction may not adequately represent all districts and years. The variation also provides motivation for comparing nonlinear models and a trend-based regression model.

4.2 Training Year Coverage

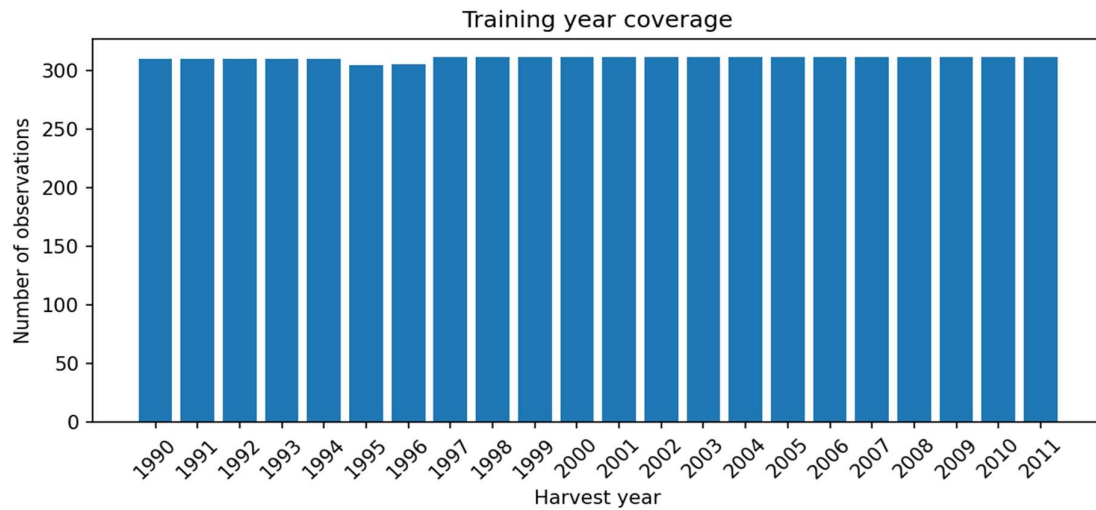


Figure: training_year_coverage.png

The year-coverage plot confirms that the training data spans the historical period from 1990 through 2011.

Interpretation : The chronological coverage is important because the objective is to evaluate how models perform when predicting later years. Keeping 2012–2015 outside the training set provides a more realistic temporal evaluation than a random split.

5. DATA PREPROCESSING

The preprocessing was implemented inside the machine-learning pipeline.

Categorical variables

The following categorical variables were one-hot encoded:

- state
- district
- season

The encoder used:

```
OneHotEncoder(handle_unknown='ignore')
```

Numerical variable

year was processed using:

1. Median imputation
2. Standard scaling

The preprocessing was fitted only on the training data and subsequently applied to validation and test observations.

This prevents information from the held-out datasets from influencing the preprocessing stage.

6. [Models Used](#)

Four regression approaches were compared.

6.1 Median Baseline

The baseline predicts the training-set median yield for every observation.

It provides a simple reference against which the machine-learning models can be evaluated.

6.2 Ridge Trend Model

Ridge regression was implemented with:

- $\alpha = 1.0$

The standardized year variable allows the model to capture an overall temporal trend while categorical variables capture regional and seasonal differences.

6.3 Decision Tree

The regression tree configuration was:

- $\text{max_depth} = 6$
- $\text{min_samples_leaf} = 10$
- $\text{random_state} = 42$

The depth limitation controls model complexity and reduces excessive fitting to individual observations.

6.4 Random Forest

The Random Forest configuration was:

- $n_estimators = 60$
- $\text{max_depth} = 12$
- $\text{min_samples_leaf} = 5$
- $n_jobs = 2$
- $\text{random_state} = 42$

The forest combines multiple randomized trees and averages their predictions to reduce variance.

7. [Validation Results](#)

Model selection was performed using **validation MAE**.

Model	Validation MAE	RMSE	R ²	Runtime (s)
Ridge Trend	0.4282	0.5583	0.7460	0.1725
Random Forest	0.5289	0.6548	0.6505	1.8251
Decision Tree	0.7160	0.8620	0.3944	0.1294
Median	0.9780	1.1902	-0.1545	0.1271

Validation Interpretation

- The Ridge Trend model achieved the lowest validation MAE of 0.4282 t/ha and was therefore selected as the final model.
- Random Forest was the second-best model with a validation MAE of 0.5289 t/ha.
- The Decision Tree performed better than the median baseline but was considerably worse than Ridge and Random Forest.
- The median baseline produced the highest validation MAE of 0.9780 t/ha.
- Therefore, the validation results indicate that the temporal trend represented by the Ridge model was particularly useful for this dataset.

8. [Model Selection](#)

The selection rule was predefined as:

Select the model with the minimum validation MAE.

Based on the validation results:

Selected model: Ridge Trend

The test data was not used during this selection.

The saved selection record confirms:

- **Selected model: ridge_trend**
- **Validation years: 2012, 2013**
- **Test years: 2014, 2015**
- **Features: state, district, season, year**
- **Target: yield_t_ha**

The selected model remained trained on the training data only and was not refitted using the validation data.

9. [Locked Test Evaluation](#)

After model selection, the final test set was evaluated.

Model	Test MAE	Test RMSE	Test R ²
Ridge Trend	0.4332 t/ha	0.5707 t/ha	0.7596
Median baseline	1.0095 t/ha	1.2346 t/ha	-0.1248

- The Ridge model achieved a test MAE of **0.4332 t/ha**, compared with **1.0095 t/ha** for the median baseline.
- This means that the selected model substantially reduced absolute prediction error compared with the simple baseline.
- The test evaluation was performed after model selection and the test lock was created before final scoring.

10. [Actual vs Predicted Analysis](#)

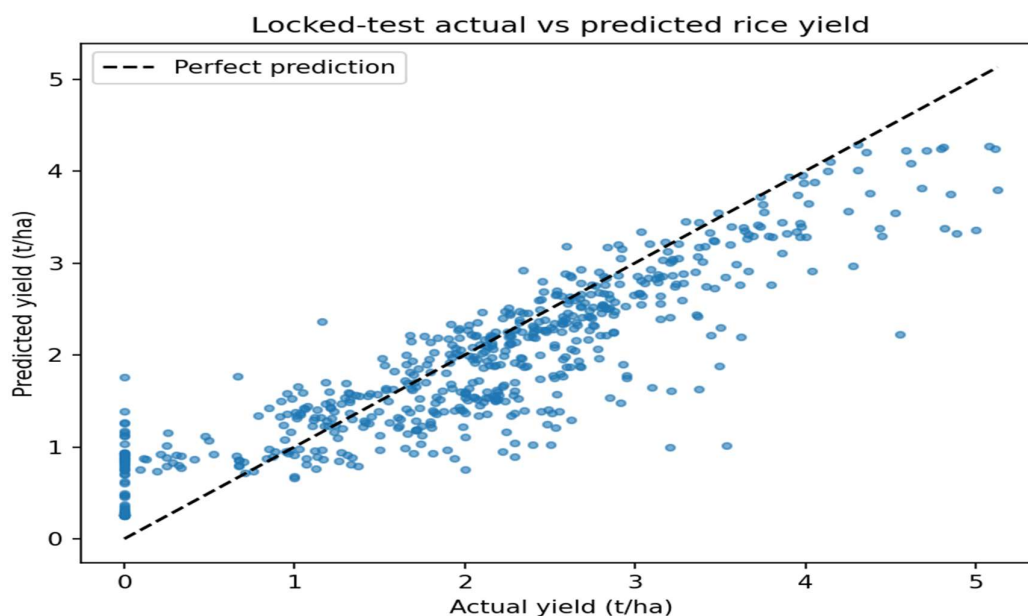


Figure: actual_predicted.png

The actual-versus-predicted plot compares observed rice yields with the predictions generated by the selected Ridge Trend model.

Interpretation : Points closer to the identity line represent more accurate predictions. The model follows the general yield pattern reasonably well, although several observations show substantial deviation from the identity relationship. The largest deviations correspond to district-year observations where the model underestimates or overestimates the actual yield.

11. [Residual Analysis](#)

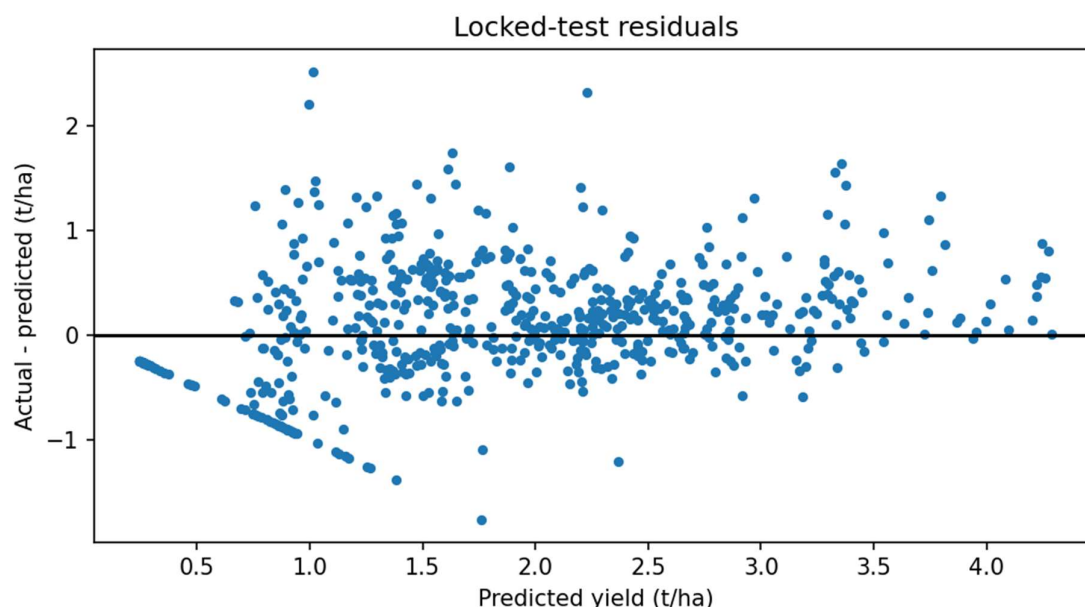


Figure: residuals.png

Residuals were calculated as:

$$e_i = y_i - \hat{y}_i$$

A positive residual represents **underprediction**, while a negative residual represents **overprediction**.

Interpretation : The residual plot shows that the model does not perfectly explain all district-level variation. Some observations have substantially larger residuals than the majority of observations. These errors may reflect factors that are not included in the predictor set, such as weather conditions, irrigation, crop variety, reporting differences or other agricultural conditions. These should be treated as possible explanations rather than confirmed causes.

12. [Five Largest Prediction Errors](#)

The five largest test errors were obtained automatically.

Row ID	State	District	Year	Actual	Prediction	Absolute Error
37 2015	Madhya Pradesh	Sehore	2015	3.5355	1.0129	2.5225
27 2015	Madhya Pradesh	Bhind	2015	4.5550	2.2274	2.3276
37 2014	Madhya Pradesh	Sehore	2014	3.2028	0.9972	2.2056
138 2014	Gujarat	Surendranagar	2014	0.0000	1.7621	1.7621
909 2014	Bihar	Bhagalpur	2014	3.3770	1.6302	1.7468

Error Interpretation

- **Sehore, 2015:** The model substantially underpredicted the observed yield. A possible explanation is district-specific variation that cannot be represented adequately using only state, district, season and year.
- **Bhind, 2015:** The model predicted 2.2274 t/ha while the actual yield was 4.5550 t/ha. This indicates a large underprediction.

- **Sehore, 2014:** A large underprediction also occurred for Sehore in 2014. The repeated occurrence suggests that district-specific effects may not be fully captured by the fitted model.
- **Surendranagar, 2014:** The actual yield was recorded as 0 while the model predicted 1.7621 t/ha. This represents an overprediction and may warrant checking the underlying agricultural record and target definition.
- **Bhagalpur, 2014:** The model underpredicted the observed yield by approximately 1.7468 t/ha.

These possible causes should not be interpreted as confirmed agricultural explanations because variables such as irrigation, variety and detailed weather information were not included.

13. [Temporal Robustness](#)

Three development-year rolling origins were evaluated.

Origin	Model	Training Maximum Year	MAE
2011	Ridge Trend	2010	0.3930
2012	Ridge Trend	2011	0.4128
2013	Ridge Trend	2012	0.4213

- The Ridge model maintained relatively stable performance across the three rolling origins.
- Its MAE increased gradually from approximately **0.3930** at the 2011 origin to **0.4213** at the 2013 origin.
- This provides descriptive evidence that the model's performance was reasonably stable across the development period, although three rolling origins are not sufficient to establish statistical confidence intervals.

14. [Year-wise Test Robustness](#)

Test Year	Number of observations	MAE	RMSE	R ²
2014	310	0.4593	0.6035	0.7304
2015	310	0.4070	0.5360	0.7866

Interpretation

- The model performed better in **2015** than in 2014 based on MAE.
- The 2014 MAE was **0.4593 t/ha**, while the 2015 MAE was **0.4070 t/ha**.
- Neither test year dominates the result excessively because both years contain the same number of observations. The difference indicates some year-to-year variation in prediction difficulty.

15. [Reproducibility and Acceptance Checks](#)

The experiment saved the model and prediction artifacts required for reproducibility.

The acceptance record confirms:

Check	Result
Split disjoint	Passed
Model reload consistency	Passed
Valid predictions	Passed
Test rows	620
Selected model	Ridge Trend

The selected model was saved as:

`outputs/core/models/selected_bundle.joblib`

The dataset hash was also stored in the model-selection record to verify that the data used during evaluation had not changed.

The test lock was created before the final test evaluation.

16. [Limitations](#)

- The results should be interpreted within the scope of the experiment.
- First, the model uses only **state, district, season and year**. Important agricultural variables such as rainfall, irrigation availability, soil properties, crop variety, fertilizer application and weather conditions are not included.
- Second, the dataset represents district-level observations rather than individual farms. Therefore, the results should not be interpreted as farm-level predictions.

- Third, the chronological split evaluates later years but does not establish generalization to completely unseen districts.
- Fourth, historical agricultural statistics may be subject to source revisions and changes in reporting practices.
- Fifth, the error analysis identifies possible hypotheses but cannot establish the actual causes of individual prediction errors.
- Finally, the model should not be interpreted as an agronomically optimal crop-selection system. Real agricultural decisions require additional agronomic, economic and environmental evidence.

17. CONCLUSION

For the evaluated district-level rice prediction scope, the **Ridge Trend model** achieved the best validation performance with a validation MAE of **0.4282 t/ha** and was selected without using the final test data.

On the locked test set, the Ridge model achieved an MAE of **0.4332 t/ha**, RMSE of **0.5707 t/ha**, and R^2 of **0.7596**, substantially outperforming the median baseline, which obtained a test MAE of **1.0095 t/ha**.

The rolling-origin results also showed relatively stable Ridge performance across the development years. However, several large district-level errors remain, demonstrating that the available predictors do not capture all agricultural variation.

Therefore, the experiment supports the conclusion that the Ridge Trend model provides a useful predictive baseline for the defined historical district-level rice-yield task, but the results **do not establish operational forecasting accuracy, spatial generalization to unseen districts, or agronomic crop-selection capability**.

Dataset References

1. Regression Dataset – ICRISAT District-Level Data

Mohapatra, S. (2023). *ICRISAT District-Level Data: Heterogeneous Climate Effect on Crop Yield and Associated Risks to Water Security in India*. Mendeley Data, Version 1.

DOI: 10.17632/yp3y5j9vv.1

License: CC BY 4.0

Data access date: 14th September 2026

2. Classification Dataset – Crop Recommendation Dataset

Ingle, A. *Crop Recommendation Dataset*. Kaggle.

The dataset contains agricultural and environmental attributes including **N, P, K, temperature, humidity, pH, rainfall**, and the corresponding **crop label**.

License: Apache License 2.0

Data access date: 14th September 2026

GITHUB REPO LINK:

https://github.com/srijithramesh24/24MID0009_Lab08_Agricultural-Predictive-Analytics

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